

PMD · Project Roadmap Architecture

Artifact #18 · v2.1 · May 2026 · 8 Phases · 66+ Projects · PMD System · Internal

ARCHITECTURE INTENT

What this document is	The master project sequencing architecture for the PMD development journey — v2.1, synthesized from multiple source frameworks. Eight progressive phases, 66+ scoped projects, a recommended reading curriculum, and offline practice assignments. The roadmap does not end at P66 — it extends as far as needed to reach portfolio milestone readiness.
What this document is not	A build guide. Projects are scoped by what they accomplish and what tools they use. How each project is built is determined project-by-project inside its own nine-phase PMD process cycle. No phase is skipped. No project closes without a retrospective.
Single evolving product strategy	From Phase III onward, projects build on a single evolving SaaS-style product codebase rather than disconnected apps. Each project adds one capability layer to the same product. By Phase VIII the cumulative product is a production-grade SaaS that becomes the flagship portfolio showcase.
User authentication placement	Authentication appears at three explicit points: P15 (Phase II) — managed auth via Supabase/Firebase, fastest path to a working login system; P19 (Phase III) — custom JWT auth from scratch, full ownership of every token, cookie, and reset flow; P25 (Phase IV) — org-level RBAC extending auth into team workspaces and invitation flows. Auth is not buried — it is an explicit named project at each level.
Portfolio milestone logic	Eight portfolio milestone candidates are marked throughout the roadmap with a star symbol (orange). No project is pre-designated as portfolio-worthy in advance. Portfolio status is earned by demonstrating PM depth, technical execution quality, and evaluator-relevant problem framing. The coach assesses readiness. The threshold emerges from the quality of the work — not from a date on a calendar.
Coaching calibration	Phases I–II: Guided Drafting — user drafts first, coach critiques. Phases III–IV: Critical Collaboration — assumptions challenged openly. Phases V–VI: CPO Peer Review — coach responds as a CPO reviewing PM work. Phases VII–VIII: Professional Friction — open disagreement, user defends every decision. Draft-first protocol is active from P1.

PORTFOLIO MILESTONE MAP

Project	Portfolio Type	Phase	Why It Works for Evaluators
P5	Product Teardown & Redesign	I	PM thinking fully visible before any backend code. Low technical dependency, high product instinct signal.
P33	Cross-Functional GTM — Pricing Launch	V	Pricing requires engineering, marketing, sales, and legal alignment. Full stakeholder coordination artifacts.
P36	Metric Optimization — Activation	V	Before/after activation metrics, funnel analysis, and A/B experiment design. Clearest PM impact story.
P43	GTM — Integrations Platform Launch	VI	Platform and ecosystem strategy. Integration catalog, OAuth2 flows, webhook reliability spec, GTM kit.

P50	0-to-1 Greenfield — AI PRD Assistant	VII	Solves a problem hiring managers experience personally. Full 0-to-1 cycle with AI, PRD, and launch plan.
P64	Analytics PM — Executive Dashboard	VIII	Business intelligence, cohort analysis, and KPI storytelling at executive reporting depth.
P65	Cross-Functional GTM — Pricing Migration	VIII	Enterprise GTM kit, stakeholder alignment artifacts, pricing migration risk log, and post-launch loop.
P66	Flagship Capstone — Full SaaS Product	VIII	Cumulative synthesis of all prior capabilities. Production SaaS with complete five-part case study suite.

USER AUTHENTICATION — WHERE IT LIVES IN THE ROADMAP

P15 · Phase II	MANAGED AUTH — Supabase or Firebase. Fastest path to a working login. Signup, login, logout, protected routes, email verification, password reset, and user-scoped data isolation. Recommended entry point: auth is live and functional before any custom backend work begins.
P19 · Phase III	CUSTOM JWT AUTH — Built from scratch. bcrypt password hashing, JWT access tokens, refresh token rotation, HTTP-only cookie storage, protected route middleware, and a full password reset flow via SendGrid email token. Full ownership — every security decision is explicit and documented.
P25 · Phase IV	ORG-LEVEL RBAC — Team and workspace layer. Workspace creation, email invitation with token validation, org-level roles (owner, admin, member), workspace switching, and member management UI. Extends P19 custom auth into a full organizational identity hierarchy.

PHASE OVERVIEW

Phase	Title	Range	Projects	Core Capabilities Introduced
I	FOUNDATION	P1 – P7	7	HTML · CSS · JS · Git · Figma · Netlify · Product communication and PRD habit from project one
II	FRONT-END FRAMEWORK	P8 – P16	9	React · TypeScript · Next.js · Headless CMS · Managed Auth (Supabase/Firebase) · Component architecture · Testing
III	BACKEND & DATABASE	P17 – P24	8	Node.js · Express · PostgreSQL · Prisma · Custom JWT Auth · File uploads · Transactional email · MongoDB
IV	PRODUCT COLLABORATION & TEAM FEATURES	P25 – P30	6	Multi-user workspaces · Team invitations · WebSockets (real-time) · Notifications · Feedback portal · Search
V	PAYMENTS, MONETIZATION & GROWTH ANALYTICS	P31 – P40	10	Stripe payments · Subscription billing · Feature flags · Behavioral analytics · A/B testing · Lifecycle email · SQL analytics · CDP
VI	REAL-TIME, SCALE & ADVANCED INFRASTRUCTURE	P41 – P47	7	BullMQ · Redis caching · OAuth2 integrations · GTM enablement · WCAG 2.2 AA audit · i18n/l10n · PWA
VII	AI & MACHINE LEARNING INTEGRATION	P48 – P56	9	OpenAI/Anthropic APIs · Vector databases · RAG pipelines · Document analysis · AI chatbots · ML model serving · Agentic workflows
VIII	PLATFORM, MOBILE & PRODUCTION SCALE	P57 – P66+	10	GraphQL · React Native mobile · Docker · CI/CD pipelines · Multi-tenant SaaS · Micro-frontends · Observability · SaaS capstone

Build the bedrock. Master HTML, CSS, JavaScript, Git, and deployment before touching any framework or library. Establish the product communication habit from day one: every project ships with a PRD, a problem statement, and at least one PM artifact. Introduce Figma for wireframing early so that design decisions precede build decisions on every project from here forward. The goal of this phase is not to build impressive products — it is to build the habit of shipping, documenting, and reflecting. Every project deploys to a live URL via the CI/CD pipeline established at P1. PM skills introduced: problem framing, persona writing, basic PRD drafting, case study structure, value proposition design, and the five-part portfolio narrative that governs every future showcase project.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Personal Link-in-Bio / Profile Page				
P1	Single-page static site: profile section, curated links, social icons, and a goal statement. Fully responsive. First GitHub commit and Netlify deploy.	Problem framing · Single-user persona · Definition of done · First PRD	HTML5 semantic structure · CSS Flexbox · CSS custom properties · Git init and commit · Netlify CI/CD pipeline	HTML5 · CSS3 · Git · GitHub · Netlify	PRD v1 · Persona · Deployed URL
	■ <i>First ship. Prove the CI/CD pipeline works end to end. Nothing else matters yet.</i>				
	Multi-Page Product Marketing Site				
P2	Multi-page marketing site for a fictional SaaS product: homepage, features, pricing, FAQ, and contact. Navigation, reusable layouts, and product messaging copy written from scratch.	Information architecture · Value proposition design · Pricing page structure · Navigation UX · Landing page conversion thinking	CSS Grid · Multi-page site structure · Tailwind CSS introduction · Vanilla JS modals and nav toggle	HTML5 · CSS3 · Tailwind CSS · Vanilla JS · GitHub · Netlify	IA diagram · PRD · Figma wireframe (intro)
	Interactive Product Calculator / ROI Estimator				
P3	Browser-based calculator — ROI estimator, pricing calculator, or productivity impact tool. User inputs, conditional logic, and results display with visual feedback. No backend.	Outcome-based product design · Input validation spec · Business logic documentation · User feedback state design	DOM manipulation · Forms and input validation · Client-side conditional logic · Basic JS error handling	Vanilla JS · HTML5 · CSS3 · GitHub	Business logic spec · PRD · Edge case log
	■ <i>Products communicate value through measurable outcomes. This project internalizes that principle.</i>				
	Responsive Case Study Template				
P4	Reusable HTML/CSS case study page template with five structured sections: problem, research, solution, metrics, tradeoffs, and learnings. This becomes the format used for every future portfolio project.	Five-part case study structure · Content hierarchy · PM artifact presentation · Visual storytelling discipline	Responsive layout · Print-friendly CSS · Semantic HTML sectioning · Typography system · Figma layout mockup	HTML5 · CSS3 · Figma · GitHub · Netlify	Case study template · Figma mockup · PRD
	Product Teardown & Redesign Case Study				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	★ PORTFOLIO CANDIDATE — Product Teardown & Redesign				
P5	Select a widely-used app with visible UX friction. Conduct a heuristic evaluation, document the friction points, design a data-backed redesign in Figma, and build a static HTML prototype of the improved flow.	UX critique methodology · Heuristic evaluation · Competitive analysis · Problem validation · OKR definition for design changes · Success metric design	Figma high-fidelity wireframes · Component-driven design · Static HTML prototype · axe-core accessibility audit intro	Figma · HTML5 · CSS3 · axe-core · GitHub	UX audit report · Figma mockup · Redesign PRD · OKRs · Case study
	■ <i>First portfolio milestone candidate. PM thinking is fully visible before a single line of backend code exists.</i>				
	Public API Data Dashboard				
P6	Dashboard consuming a public REST API (weather, finance, GitHub, movies). Multiple chart types, loading states, error states, and empty states handled explicitly. No backend required.	API dependency risk documentation · Data visualization principles · Dashboard UX hierarchy · Third-party SLA awareness	Fetch API · Async/Await · REST API consumption · JSON parsing · Chart.js · Error and loading state patterns	Vanilla JS · Fetch API · Chart.js · OpenWeatherMap or CoinGecko API · GitHub	API integration spec · Error state matrix · Dashboard wireframe · PRD
	Multi-Page Portfolio Site with Vanilla JS Routing				
P7	Rebuild thepmdev.com foundation as a content-rich multi-page site with JS-driven navigation, animated page transitions, dynamic content injection, GA4, and Open Graph meta tags.	IA design · Navigation strategy · SEO fundamentals · Content hierarchy · Personal brand positioning	History API / hash routing · JS module pattern · CSS transitions · Meta and OG tags · GA4 snippet integration	Vanilla JS modules · History API · CSS transitions · Google Analytics 4 · GitHub · Netlify	IA diagram · Navigation spec · SEO checklist · PRD
	■ <i>Closes Phase I. Before touching React, understand what a framework is actually solving.</i>				

Graduate from document-oriented thinking to component-oriented thinking. React is the dominant front-end framework in product companies — master it deeply before adding a custom backend. This phase introduces headless CMS for content management and, critically, managed user authentication via Supabase or Firebase at P15. Managed auth is the fastest path to a working login system and the recommended entry point for user accounts. Custom auth from scratch is built in Phase III (P19). PM skills introduced: design systems, component spec writing, content modeling, migration planning, identity lifecycle design, and A/B hypothesis framing.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	React Component Library & Design System				
P8	Reusable component library: Button, Input, Card, Modal, Toast, Badge, Dropdown. Each component documented with props, states, and usage examples. Pure UI primitives — no app logic.	Component spec writing · Design token documentation · Reusability criteria · Accessibility requirements per component	React functional components · Props and PropTypes · CSS Modules · TypeScript introduction · Vite project setup	React · Vite · CSS Modules · TypeScript · axe-core	Component spec · Design token inventory · Accessibility checklist · PRD
	React Task Board (Kanban) with State Management				
P9	Trello-style Kanban board: create, edit, delete, and move tasks between columns. Tag filtering, priority levels, due dates, and LocalStorage persistence across sessions.	CRUD interaction design · Prioritization logic · Data model definition · Workflow design	useState · useReducer · useEffect · HTML Drag-and-Drop API · Conditional rendering · Component composition · LocalStorage	React · Vite · TypeScript · Tailwind CSS · LocalStorage	Data model spec · Interaction spec · PRD
	■ Kanban is the most PM-relevant interface pattern. Build it and understand it from the inside.				
	Product Requirements Manager App				
P10	Web app for writing lightweight PRDs: problem statement, user personas, requirements, success metrics, risks, and open questions. Structured forms, Markdown editor, and template system.	PM artifact design · Structured data modeling · Reusable template patterns · Information hierarchy in forms	React controlled forms · Markdown editor integration · Template pattern · TypeScript interfaces · LocalStorage schema	React · TypeScript · Tailwind CSS · Marked.js or react-md-editor · LocalStorage	PRD template spec · Data model · PRD
	■ A PM building a PRD tool understands the artifact from the inside out.				
	Now / Next / Later Roadmap Builder				
P11	Roadmap planning app: create initiatives, assign RICE or MoSCoW priority scores, estimate effort, group by Now/Next/Later horizon, and export as image or PDF for stakeholder sharing.	RICE framework · MoSCoW framework · ICE scoring · Roadmap structure · Tradeoff communication · Stakeholder presentation design	React Router v6 · Dynamic route params · Recharts or Chart.js timeline · Client-side PDF/image export · TypeScript	React · React Router · Recharts or Chart.js · TypeScript · Tailwind CSS	Prioritization framework comparison doc · Roadmap spec · Export spec · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	■ <i>Use RICE to prioritize this project's own feature backlog. PM exercise embedded in the build.</i>				
	Product Metrics Dashboard Prototype				
P12	Dashboard showing product KPIs: signups, activation rate, retention, churn, conversion, and MRR. Mock JSON data. Multiple chart types. AARRR funnel visualization with drop-off callouts.	KPI definition · AARRR funnel design · Dashboard layout hierarchy · Metric interpretation · Data storytelling	Recharts · Custom hooks · Memoization · Skeleton loading states · TypeScript generics · Mock data structure	React · Recharts · TypeScript · Tailwind CSS · Mock JSON	KPI definition doc · Dashboard spec · AARRR map · PRD
	■ <i>Foundation for every future analytics case study. Metrics defined here reappear across the journey.</i>				
	Global State Management — Shopping Cart				
P13	Product listing, cart, and simulated checkout flow. Add/remove/update quantities, persist cart to LocalStorage, and global state via Zustand. Simulate payment confirmation UI.	Cart UX patterns · State persistence strategy · Checkout flow design · Payment failure state UX · Plan upgrade/downgrade analogy	Zustand or Redux Toolkit · Context API comparison · Selector patterns · Middleware · Optimistic UI	React · Zustand or Redux Toolkit · TypeScript · Tailwind CSS	State architecture diagram · Cart flow spec · PRD
	Next.js SSR/SSG Content Site with Headless CMS				
P14	Content-driven site powered by a headless CMS (Contentful or Sanity). Static generation for performance, dynamic pages where needed, SEO-optimized meta tags, and image optimization.	Content modeling · SEO strategy · Static vs dynamic routing tradeoffs · Non-technical stakeholder tooling design	Next.js · Static Site Generation · Server-Side Rendering · Contentful or Sanity · next/image · next/head SEO	Next.js · Contentful or Sanity · TypeScript · Tailwind CSS · Vercel	Content model spec · SEO checklist · Static vs dynamic decision doc · PRD
	User Accounts & Authentication — Managed (Supabase/Firebase)				
	■ AUTH MILESTONE				
P15	Full user account system via managed cloud auth: signup, login, logout, protected routes, password reset, email verification, and user-scoped data isolation. First working login in the product.	Auth flow documentation · Identity lifecycle design · Onboarding trust signals · Security requirement writing · Account lifecycle (create, verify, reset, delete)	Supabase Auth or Firebase Auth · Protected routes in Next.js · Row-level security intro · Real-time DB subscriptions · Email verification flow	Supabase or Firebase · Next.js · TypeScript · Tailwind CSS	Auth flow diagram · Identity lifecycle doc · Security requirements · PRD
	■ AUTH MILESTONE — <i>Earliest point in the roadmap where full user accounts are live. Managed service removes infrastructure burden so focus stays on the product experience.</i>				
	Component Testing with Jest and React Testing Library				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P16	Full test suite for P9 or P13. Unit tests per component, integration tests for key user flows, coverage report generated and interpreted, and a CI test run on GitHub Actions.	Acceptance criteria as test cases · QA strategy documentation · Release confidence definition · CI gate criteria	Jest · React Testing Library · Test coverage tooling · Mocking · GitHub Actions CI pipeline introduction	Jest · React Testing Library · GitHub Actions · TypeScript	Test strategy doc · Coverage report · CI pipeline spec · PRD
	■ Closes Phase II. Acceptance criteria traced directly to tests. A PM who understands this writes better requirements.				

Cross the full-stack line. Build custom APIs, connect to real relational and document databases, and implement authentication from scratch — giving complete ownership of every layer of the stack. P19 is the custom JWT auth project: bcrypt hashing, refresh token rotation, HTTP-only cookies, and full password reset. This is the phase where the product transitions from prototype to a real, deployable full-stack application. File uploads, cloud storage, and transactional email are introduced as production-grade product capabilities. PM skills introduced: API contract design, data modeling, ERD documentation, technical requirement writing, system architecture diagramming, and migration planning.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Custom REST API with Node.js and Express				
P17	Standalone REST API for the P9 Task Manager. Full CRUD endpoints, request validation middleware, error handling, rate limiting, and a complete Postman collection with documentation.	API contract documentation · Endpoint design · HTTP status code strategy · Versioning rationale · Rate limit requirement	Node.js · Express.js · RESTful routing · Middleware chain · Postman collection · dotenv · express-validator	Node.js · Express.js · Postman · dotenv · express-validator	API contract doc · Postman collection · Rate limit spec · PRD
	PostgreSQL Database Integration with Prisma ORM				
P18	Replace LocalStorage with a custom PostgreSQL database. Schema design, Prisma migrations, relational queries, indexed lookups, and a formal ERD documenting all entity relationships.	Data model documentation · Migration planning · Relational vs flat data tradeoffs · Schema versioning strategy	PostgreSQL · Prisma ORM · Database migrations · Schema design · SQL fundamentals · Indexed queries	PostgreSQL · Prisma · Node.js · Express.js · pgAdmin	ERD (Entity-Relationship Diagram) · Migration log · PRD
	■ SQL is a non-negotiable PM hard skill. Every data-fluent PM can write a basic SELECT query.				
	Custom User Auth — JWT, Refresh Tokens & Password Reset				
	■ AUTH MILESTONE				
P19	Full custom authentication system built from scratch: email/password registration, bcrypt hashing, JWT access tokens, refresh token rotation, HTTP-only cookies, protected routes, and password reset via email token.	Custom auth flow documentation · Session UX design · Security requirement writing · Trust and privacy spec · Token lifecycle policy	JWT · bcrypt · Refresh token rotation · HTTP-only cookies · Role-based route protection · SendGrid password reset email	JWT · bcrypt · Node.js · Express.js · PostgreSQL · SendGrid	Auth flow diagram · Token lifecycle doc · Security requirements · PRD
	■ AUTH MILESTONE — Full-ownership custom auth. Every decision is explicit: token expiry, rotation, cookie policy, and reset flow.				
	Full-Stack Integrated App — React + Node + PostgreSQL				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P20	First fully integrated full-stack application. React frontend consumes the custom authenticated Node/Express/PostgreSQL API. Users register, log in, and manage their own data end to end.	Full system architecture documentation · Cross-layer dependency mapping · End-to-end user journey · Deployment runbook	Full-stack integration · CORS configuration · Environment variable management · Frontend + backend co-deployment · Render or Railway hosting	React · Node.js · Express · PostgreSQL · Prisma · Render or Railway	System architecture diagram · Full-stack PRD · Deployment runbook
	■ <i>First project where every layer is yours. The debugging will teach more than any tutorial.</i>				
	Role-Based Access Control (RBAC) + Admin Dashboard				
P21	Extend P20 with admin and standard user roles. Admin dashboard with user management, content moderation, system settings, and an audit log of all admin actions. Permission-gated UI components.	Permission matrix design · Admin UX patterns · Internal tooling design · Audit trail requirements · Operational visibility	RBAC middleware · Permission-gated rendering · Audit logging pattern · Admin UI · Soft delete	Node.js · PostgreSQL · React · JWT with role claims · Prisma	Permission matrix · Admin flow spec · Audit log spec · PRD
	File Uploads, Cloud Storage & Attachments				
P22	Users upload files and images as task attachments. Cloud storage via AWS S3 or Supabase Storage. File type and size validation, presigned URLs for secure access, and download/preview UI.	File handling UX · Storage cost modeling · Upload failure state design · File lifecycle and retention policy	AWS S3 or Supabase Storage · Multer · Presigned URLs · File validation middleware · Client-side preview	AWS S3 or Supabase Storage · Multer · Node.js · React	File handling spec · Storage cost estimate · Lifecycle policy · PRD
	Transactional Email System with SendGrid				
P23	Transactional emails: welcome on registration, password reset, task assignment notification, weekly digest, and account deletion confirmation. HTML templates, delivery tracking, and bounce handling.	Transactional email spec · User communication design · Deliverability requirements · Email metrics: open, bounce, unsubscribe	SendGrid API · HTML email templating · Webhook handling for delivery events · Unsubscribe management · Email preview testing	SendGrid · Node.js · Express.js · HTML email templates · Prisma	Email flow spec · Template inventory · Delivery metrics spec · PRD
	■ <i>Every automated message is a product decision. Communication design is a PM craft.</i>				
	NoSQL Document Store — Activity Feed with MongoDB				
P24	Activity feed and event log system using MongoDB. Flexible document schema, pagination, filter and sort by event type. Structured SQL vs NoSQL tradeoff analysis documented as a PM artifact.	SQL vs NoSQL decision framework · Schema flexibility cost-benefit · Data architecture documentation	MongoDB · Mongoose · Document schema design · Aggregation pipeline · Indexed queries · Cursor-based pagination	MongoDB · Mongoose · Node.js · Express.js	SQL vs NoSQL decision doc · Schema design · Aggregation query library · PRD
	■ <i>Closes Phase III. The decision doc produced here is a real PM technical artifact.</i>				

Phase IV · P25 – P30

PRODUCT COLLABORATION & TEAM FEATURES

Build the multi-user collaborative product layer — the feature set that separates a personal tool from a team product. This phase builds directly on the custom authentication established in Phase III (P19). Teams, workspaces, email invitations, and org-level role management extend the auth system into a full identity hierarchy. Real-time collaboration via WebSockets, threaded comments, a notification center, a public customer feedback portal, and full-text search complete the collaboration feature set. By the end of this phase the product is a functioning team collaboration platform. PM skills introduced: multi-tenant data modeling, team onboarding flows, permission hierarchies, customer discovery systems, voice-of-customer synthesis, and search product design.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Team Workspaces, Invitations & Org-Level Auth				
	■ AUTH MILESTONE				
P25	Users create workspaces, invite team members via email with token-validated links, and collaborate on shared projects. Org-level RBAC (owner, admin, member), workspace switching, and member management UI.	Multi-user data model design · Team onboarding flow · Invitation UX · Permission hierarchy documentation · Org-level identity design	Organization/team schema · Email invitation with token validation · RBAC at org level · Workspace context middleware · Scoped API queries	Node.js · PostgreSQL · Prisma · React · SendGrid · JWT	Team data model diagram · Invitation flow spec · Permission hierarchy matrix · PRD
	■ AUTH MILESTONE — Team and org layer. Extends P19 custom auth into a full organizational identity hierarchy.				
	Comments, @Mentions & Threaded Discussions				
P26	Threaded comments on tasks and projects. @mention notifications, emoji reactions, comment edit/delete with audit trail, and comment-level permission control based on org role.	Collaborative UX design · Notification trigger taxonomy · Mention and threading interaction spec · Comment lifecycle policy	Nested comment schema · @mention parsing · Notification trigger events · Soft delete with audit · React rich text basics	Node.js · PostgreSQL · Prisma · React · WebSockets (intro)	Comment system spec · Notification trigger map · PRD
	Real-Time Collaboration with WebSockets				
P27	Live updates pushed to all workspace members without refresh. User presence indicators, typing status, and live task state synchronization across all active users. Operational status panel.	Real-time UX spec · Presence design · Message delivery guarantee definition · Concurrent user experience mapping · Latency tolerance rules	WebSockets · Socket.io · Room-based event broadcasting · Presence tracking · Optimistic UI patterns · Redis pub/sub introduction	Socket.io · Node.js · PostgreSQL · React · Redis (pub/sub intro)	Real-time spec · Presence flow diagram · Latency tolerance rules · PRD
	In-App Notification Center				
P28	Multi-channel notification system: in-app notification bell with unread badge, email notifications per event type, user preference controls per channel, and notification analytics.	Notification taxonomy design · Opt-in/opt-out UX · Notification fatigue strategy · Multi-channel preference management spec	Notification schema · Unread count aggregation · WebSocket-driven badge updates · Per-user preference storage · SendGrid channel	Node.js · PostgreSQL · Socket.io · React · SendGrid	Notification taxonomy · Preference matrix · Fatigue analysis doc · PRD

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	■ Notification strategy is one of the most user-trust-sensitive decisions a PM makes.				
	Customer Feedback & Feature Request Portal				
P29	Public-facing portal where users submit feedback, vote on feature requests, and track status. Admins categorize, respond, change status, and link requests directly to roadmap items.	Voice-of-customer system design · Feedback synthesis methodology · Qualitative data loop · Roadmap transparency · Customer discovery documentation	Voting and ranking system · Status workflow state machine · Public vs authenticated views · Admin moderation queue · Email notification on status change	Next.js · PostgreSQL · Prisma · React · SendGrid · Auth	Feedback taxonomy · Prioritization workflow spec · Customer discovery log · PRD
	■ Strong PM relevance. The artifacts here are real discovery outputs, not simulations.				
	Full-Text Search Across Workspaces				
P30	Search across tasks, projects, comments, and documents within a workspace. Typo tolerance, faceted filtering, relevance ranking, search analytics (no-results rate, top queries), and zero-results handling.	Search relevance definition · Zero-results handling strategy · Search analytics interpretation · Discoverability UX · Search as a retention feature	Algolia or PostgreSQL full-text search · Index schema design · Relevance tuning · Faceted filtering · Search event tracking	Algolia or PostgreSQL FTS · Node.js · React · TypeScript	Search spec · Relevance criteria doc · Zero-results strategy · PRD
	■ Closes Phase IV. Search is a product strategy decision masquerading as a technical one.				

Build products that make money and measure themselves. Stripe integration, subscription lifecycle management, feature flags, behavioral analytics, A/B testing, lifecycle email automation, SQL analytics, and a complete customer data platform. This phase produces the strongest PM portfolio candidates: metric optimization projects and GTM case studies. Every project in this phase adds a measurable business capability to the product built in Phases III and IV. PM skills introduced: pricing model design, funnel analysis, activation metric definition, A/B experiment design, statistical significance, LTV/CAC modeling, event taxonomy, and CDP architecture.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Stripe One-Time Payment Integration				
P31	One-time payment processing via Stripe Checkout or Elements. Webhook handling for payment confirmation, failure, and refund events. Purchase confirmation email. Idempotency handled correctly.	Payment flow documentation · Checkout UX design · Payment failure state handling · PCI compliance awareness · Webhook event taxonomy	Stripe API · Stripe Webhooks · Idempotency keys · Payment intent lifecycle · Webhook signature verification	Stripe · Node.js · Express.js · React · SendGrid	Payment flow diagram · Webhook event spec · Failure handling matrix · PRD
	Subscription Billing — SaaS Tier Model				
P32	Full subscription lifecycle: free trial, plan selection (free/pro/team), upgrade/downgrade, cancellation, reactivation, and invoice history. Stripe Customer Portal integrated. Plan entitlement middleware enforced.	Pricing tier design · Subscription lifecycle mapping · Plan entitlement logic · Churn definition · LTV/CAC fundamentals · Billing UX edge cases	Stripe Subscriptions · Stripe Customer Portal · Billing webhooks · Entitlement middleware · Plan upgrade/downgrade API logic	Stripe · Node.js · PostgreSQL · Prisma · React	Pricing tier spec · Subscription lifecycle diagram · Entitlement matrix · Billing edge case log · PRD
	■ <i>Subscription logic is the most common PM domain in SaaS. Own every edge case completely.</i>				
	Pricing Page, Packaging Design & Upgrade Prompts				
	★ PORTFOLIO CANDIDATE — Cross-Functional GTM: Pricing Launch				
P33	Pricing comparison page with psychological pricing design, contextual in-product upgrade prompts, billing portal link, and a documented A/B test variant of pricing copy structure.	Pricing psychology · Packaging strategy · Conversion optimization · Upgrade prompt placement rationale · Plan comparison UX · Pricing page audit	Feature-gated UI components · Contextual upgrade modals · Pricing page A/B variant structure · Stripe plan switching UI	React · Stripe · Tailwind CSS · TypeScript	Pricing rationale doc · Upgrade prompt spec · A/B hypothesis · PRD
	■ <i>Portfolio candidate. Pricing decisions require engineering, marketing, sales, and legal alignment.</i>				
	Feature Flags with LaunchDarkly				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P34	Feature flags across the product: gradual rollout, kill switch, per-user targeting, percentage rollout, and beta group management. Flag inventory documented and lifecycle managed.	Rollout strategy design · Flag lifecycle management · Percentage rollout risk assessment · Beta group definition · Kill switch rationale	LaunchDarkly SDK · Flag targeting rules · A/B flag variants · Flag cleanup discipline · CI/CD integration for flag evaluation	LaunchDarkly · React · Node.js · TypeScript	Flag inventory · Rollout strategy doc · Flag lifecycle policy · PRD
	Product Analytics — Behavioral Event Taxonomy				
P35	Instrument the product with behavioral event tracking. Define and implement the full event taxonomy. Build activation funnel, retention cohorts, and feature adoption dashboard. Identify first major drop-off point.	Event taxonomy design · Funnel definition · Activation metric selection · AARRR framework application · Retention cohort analysis	Mixpanel or Amplitude SDK · Event tracking implementation · Funnel visualization · Cohort analysis · Segment.io introduction	Mixpanel or Amplitude · Segment.io (optional) · React · Node.js	Event taxonomy doc · Funnel spec · AARRR dashboard · Activation definition · PRD
	■ <i>The event taxonomy is the most important data decision a PM makes. Get it right early.</i>				
	Onboarding Flow Redesign & Activation Optimization				
	★ PORTFOLIO CANDIDATE — Metric Optimization: Activation Funnel				
P36	Multi-step onboarding wizard with progress tracking. Measure drop-off at each step using analytics. Redesign the highest-dropout step. Document before/after activation rate comparison as a PM case study.	Onboarding funnel analysis · Activation metric definition · Drop-off hypothesis generation · Iteration decision documentation · PM impact story construction	Multi-step wizard UI · Onboarding progress persistence · Event tracking per step · Feature flag for variant · Before/after data comparison	React · Node.js · Mixpanel or Amplitude · LaunchDarkly · PostgreSQL	Onboarding funnel doc · Drop-off analysis · Redesign rationale · Before/after metrics · Case study
	■ <i>Portfolio candidate. 'How I improved activation by X%' is the clearest PM impact story a candidate can tell.</i>				
	A/B Testing — Hypothesis to Statistical Significance				
P37	Design and run a controlled A/B test on a live feature (CTA copy, onboarding step, or pricing prompt). Hypothesis, sample size calculation, variant via LaunchDarkly, results analysis, and rollout/rollback decision memo.	A/B test design discipline · Statistical significance · MDE calculation · Null hypothesis framing · Rollout/rollback decision memo	LaunchDarkly variants · Chi-square or z-test basics · Sample size calculator · Results visualization in analytics platform	LaunchDarkly · Mixpanel or Amplitude · React · Statistics fundamentals	A/B test design doc · Statistical significance calculation · Results report · Rollout decision memo · PRD
	Lifecycle Email & Behavioral Retention System				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P38	Behavioral email sequences triggered by user actions: welcome, onboarding incomplete reminder, feature adoption nudge, weekly digest, reactivation campaign, and churn risk alert. Segmented by behavioral cohort.	Lifecycle messaging strategy · Retention loop design · User segmentation · Email metrics (open, click, unsubscribe, downstream conversion) · Churn prevention framework	SendGrid or Resend · Cron job scheduling · Behavioral trigger logic · BullMQ job queue · User segment SQL queries	SendGrid or Resend · BullMQ · Redis · Node.js · PostgreSQL · Cron	Lifecycle email map · Segment definitions · Retention loop diagram · Email metrics spec · PRD
	SQL Analytics — Direct Database Querying for PMs				
P39	Write SQL queries directly against the product database to answer business questions: funnel completion rates, cohort retention, MRR by plan, churn by segment, and feature adoption by role.	Data question formulation · SQL as PM self-service tool · Analyst collaboration simulation · KPI storytelling from raw data · Dashboard design	Advanced SQL · Window functions · CTEs · Aggregation queries · Metabase or Redash dashboard · pgAdmin query execution	PostgreSQL · Metabase or Redash · SQL · pgAdmin	SQL query library · Self-service KPI dashboard · Data question log · PRD
	■ A PM who writes SQL does not wait three days for an analyst to answer a question.				
	Customer Data Platform — Segment.io + Data Export				
P40	Integrate Segment.io as a central event router connecting analytics, email, and CRM. Add CSV and PDF data export for users and admin reporting. Single tracked event fires to all downstream tools simultaneously.	CDP architecture · Data governance fundamentals · Single source of truth design · Export UX · Admin reporting dashboard design	Segment.io source and destination config · Identity resolution · Event routing rules · CSV/PDF export libraries · BullMQ for async export	Segment.io · Node.js · React · Mixpanel · SendGrid · pdf-lib or similar	CDP architecture diagram · Event routing map · Export spec · Admin reporting spec · PRD
	■ Closes Phase V. Every growth-stage company uses a CDP. Understanding it makes you a better PM immediately.				

Build products that operate at concurrency and scale. Background job processing, Redis caching and performance optimization, an integrations marketplace with OAuth2 and webhook reliability, internal sales enablement as a GTM artifact, a full WCAG 2.2 AA accessibility audit, internationalization and localization, and a PWA for mobile reach. This phase produces enterprise-grade technical fluency and a second strong GTM portfolio candidate. PM skills introduced: SLA writing, latency tolerance definition, integration marketplace design, OAuth2 UX, GTM kit construction, WCAG compliance documentation, i18n product strategy, and offline-first product design.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	Background Job Queue with BullMQ and Redis				
P41	Async background processing: email delivery, report generation, image processing, and data exports. Job retry logic with exponential backoff, dead letter queue, and admin job status dashboard.	Async vs sync decision framework · Job failure UX · SLA definition for background tasks · Operational status design · Queue as product reliability layer	BullMQ · Redis · Worker processes · Retry/backoff patterns · Dead letter queue handling · Job status WebSocket updates	BullMQ · Redis · Node.js · PostgreSQL · React	Job taxonomy · Failure handling spec · SLA document · PRD
	Redis Caching Layer & Performance Optimization				
P42	Redis caching on high-read endpoints. Cache invalidation strategy, TTL design, cache-hit rate measurement, before/after latency comparison, and Google Lighthouse performance audit with Web Vitals baseline.	Performance requirement writing · Cache strategy documentation · Latency SLA definition · Page speed to conversion rate connection · Performance budget definition	Redis caching · TTL and eviction strategy · Cache invalidation patterns · Google Lighthouse automation · Web Vitals measurement	Redis · Node.js · Google Lighthouse · Postman · pgAdmin	Performance baseline report · Cache strategy doc · Web Vitals dashboard · Performance budget · PRD
	Integrations Marketplace — OAuth2 & Outbound Webhooks				
	★ PORTFOLIO CANDIDATE — GTM: Integrations Platform Launch				
P43	Integration hub connecting to external services (Slack, Google Calendar, GitHub). OAuth2 authorization flows for each integration, outbound webhooks with retry logic, and an integration management UI.	Integration product design · OAuth2 UX flow · Webhook reliability spec · Partner integration documentation · API ecosystem and platform strategy	OAuth2 authorization code flow · Token storage and refresh · Webhook payload handling and retry · Integration state management	OAuth2 · Node.js · PostgreSQL · React · Slack API or Google Calendar API	Integration catalog · OAuth2 flow diagrams · Webhook reliability spec · PRD
	■ <i>Integrations are a platform strategy decision. This project introduces ecosystem thinking to the PM toolkit.</i>				
	Internal Sales Enablement Portal				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P44	Internal-facing GTM portal: product positioning guides, objection handling scripts, competitive comparison matrices, launch FAQs, customer segment profiles, and demo scripts. Searchable and role-gated.	GTM planning · Sales enablement documentation · Stakeholder alignment artifact creation · Influence without authority · Launch readiness checklist	Next.js · Admin roles · Markdown or CMS content layer · Search integration · Authentication · PDF export	Next.js · PostgreSQL · Prisma · Auth · Markdown · Search	GTM launch kit · Sales enablement guide · Competitive comparison matrix · Launch readiness checklist · PRD
	■ These artifacts are real GTM deliverables. A hiring manager reviewing this understands cross-functional execution.				
	WCAG 2.2 AA Accessibility Audit & Full Remediation				
P45	Full WCAG 2.2 Level AA audit of the existing product. Automated audit via axe-core, manual keyboard navigation testing, screen reader testing (NVDA or VoiceOver), color contrast remediation, focus management, and ARIA implementation.	Accessibility as product requirement · WCAG 2.2 compliance documentation · Audit-to-remediation prioritization · Inclusive design principles · Legal risk framing	axe-core automated audit · ARIA roles and labels · Keyboard navigation patterns · Focus management · NVDA or VoiceOver testing · Colour Contrast Analyser	axe-core · NVDA or VoiceOver · Colour Contrast Analyser · React · Node.js	WCAG 2.2 audit report · Remediation priority matrix · Compliance sign-off checklist · PRD
	■ WCAG 2.2 AA is non-negotiable on all PMD projects. This project builds the formal audit skill.				
	Internationalization & Localization (i18n / l10n)				
P46	Multi-language support: string externalization, locale switching UI, RTL layout support (Arabic or Hebrew), date/number/currency formatting per locale, and a translation workflow using Crowdin or similar.	Localization requirement writing · Market entry documentation · Translation workflow design · RTL product considerations · Regional pricing strategy	i18next · React-i18next · RTL CSS layouts · Locale-aware number and date formatting · Pluralization rules · Translation file management	i18next · React · Node.js · Crowdin (optional) · Tailwind CSS RTL	Localization spec · Locale support matrix · Translation workflow doc · Market entry rationale · PRD
	Progressive Web App (PWA) — Mobile Bridge				
P47	Convert the existing product to a PWA: service worker for offline caching, install-to-home-screen prompt, Web Push API notifications, app-shell architecture, and a Lighthouse PWA audit.	Offline-first UX requirements · PWA vs native mobile decision framework · Push notification strategy · App install conversion design · Mobile reach without app store	Service Worker API · Cache API and strategies · Web Push API · Manifest.json · Workbox · Lighthouse PWA audit scoring	Service Worker API · Web Push API · Workbox · React · Lighthouse	Offline UX spec · PWA vs native decision doc · Push notification taxonomy · Lighthouse PWA audit · PRD
	■ Closes Phase VI. PWA is the most cost-effective mobile strategy for a solo product builder at this stage.				

Integrate AI as a deliberate product capability — not as a novelty, but as a solution to a defined user problem. OpenAI and Anthropic API integration, structured output design, vector search, RAG pipelines, document analysis, AI chatbots with memory and guardrails, ML model serving, and autonomous agentic workflows. This phase produces the strongest 0-to-1 greenfield portfolio candidates in the entire roadmap. PM skills introduced: AI feature scoping, prompt engineering as product design, hallucination risk documentation, AI cost modeling, responsible AI and content policy writing, model card documentation, and agentic UX design.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	OpenAI API — Text Generation Feature Integration				
P48	AI text generation inside an existing product: task description generation, summary creation, or meeting notes. Streaming responses, token tracking, rate limiting, cost monitoring per feature.	AI feature scoping · Prompt as product design · Cost-per-feature analysis · AI failure mode documentation · Human-in-the-loop design	OpenAI API · Streaming responses · Token budgeting · Rate limiting · Error classification · Cost monitoring dashboard	OpenAI API · Node.js · React · TypeScript	AI feature spec · Prompt library v1 · Cost model · Failure mode doc · PRD
	Anthropic Claude API — Structured Output Product				
P49	Product feature using Claude producing structured JSON for downstream consumption. Multi-turn conversation state management, system prompt engineering, output schema validation, and evaluation rubric.	Structured output spec writing · System prompt as product requirement · AI output evaluation criteria · Open-ended vs structured output tradeoffs	Anthropic SDK · System prompts · Multi-turn conversation management · JSON output schema · Zod schema validation · Output regression testing	Anthropic SDK · Node.js · React · TypeScript · Zod	System prompt spec · Output schema · Evaluation rubric · Regression test suite · PRD
	AI PRD Assistant — Greenfield 0-to-1 Product				
	★ PORTFOLIO CANDIDATE — 0-to-1 Greenfield: AI PRD Assistant				
P50	AI assistant that turns vague product ideas into structured PRDs: problem statement, personas, user stories, success metrics, risks, and open questions. Human-in-the-loop editing and versioned output history.	Full PM lifecycle ownership · Product-market fit validation · AI UX design · Prompt engineering as requirement · 0-to-1 product launch plan	OpenAI or Anthropic API · Streaming · Structured output parsing · Version history in PostgreSQL · React rich text editing	Anthropic or OpenAI API · Next.js · PostgreSQL · Prisma · Stripe · TypeScript	Market research · Personas · Full PRD · RICE prioritization · Launch plan · Full case study
	■ <i>Strongest greenfield candidate. Solves a problem hiring managers recognize and experience personally.</i>				
	Vector Database & Semantic Search				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P51	Embed a document corpus and build semantic search over it. Similarity scoring, hybrid search combining keyword and semantic retrieval, relevance evaluation, and embedding model comparison.	Semantic vs keyword search tradeoffs · Relevance evaluation design · Embedding cost analysis · Search quality metrics	Pinecone or pgvector · OpenAI Embeddings API · Cosine similarity · Hybrid search implementation · Chunking strategy experiments	Pinecone or pgvector · OpenAI Embeddings · Node.js · TypeScript	Search architecture diagram · Relevance evaluation doc · Embedding cost model · Chunking strategy · PRD
	RAG Pipeline — Retrieval-Augmented Generation				
P52	RAG system over a document corpus: ingest, chunk, embed, retrieve, and generate. Document Q&A; with source citations, hallucination mitigation strategy, and context window management experiments.	RAG architecture documentation · Hallucination risk spec · Citation UX design · Grounding strategy · Context window tradeoffs	LangChain or custom RAG pipeline · Vector retrieval · Context window management · Source citation injection · Chunking strategy iteration	LangChain · Pinecone or pgvector · OpenAI or Anthropic · Node.js · TypeScript	RAG architecture diagram · Hallucination mitigation spec · Chunking strategy doc · PRD
	AI Document Analysis Tool				
P53	Upload a PDF or contract. AI extracts structured data (key clauses, dates, parties), generates a summary, answers specific questions, and flags risks. Multi-document side-by-side comparison mode.	Document processing UX · Extraction accuracy definition · Edge case documentation · Confidence scoring design · Human-review trigger design	PDF text extraction · Multimodal input handling · Structured extraction prompts · Confidence scoring · pypdf or pdfplumber	Anthropic or OpenAI API · pypdf · Node.js · React · TypeScript	Extraction spec · Accuracy evaluation rubric · Edge case log · PRD
	AI Chatbot with Memory, Persona & Guardrails				
P54	Persistent AI assistant with conversation memory, defined persona, tool use (web search or calculator), content guardrails, and a structured user feedback loop for quality improvement.	Persona specification · Guardrail requirement writing · Content policy documentation · Feedback loop design · Trust and safety spec	Conversation memory patterns · Tool use / function calling · Content filtering logic · Feedback storage · Conversation history in PostgreSQL	Anthropic or OpenAI API · LangChain or custom · PostgreSQL · React · TypeScript	Persona spec · Guardrail spec · Content policy doc · Feedback loop design · PRD
	ML Predictive Feature — Churn Prediction or Recommendation				
P55	Train and deploy a simple ML model as a product feature: churn prediction or content recommendation. Feature engineering, model evaluation, A/B test against rule-based baseline, and model card documentation.	ML feature requirements writing · Model evaluation criteria · False positive/negative cost analysis · Explainability requirements · Model card as PM artifact	Python · scikit-learn · Feature engineering · Model evaluation metrics (AUC, F1, precision/recall) · FastAPI model serving endpoint	Python · scikit-learn · pandas · FastAPI · PostgreSQL	ML feature spec · Model card · A/B test design · Performance metrics report · PRD
	■ <i>First Python project. PM must understand model tradeoffs deeply to write good ML requirements.</i>				
	AI Agent with Tool Orchestration				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P56	Autonomous agent that plans, selects tools (web search, code execution, API calls), and completes multi-step tasks with minimal human input. Full observability, tracing, and failure recovery built in from the start.	Agentic product design · Human-in-the-loop decision points · Observability requirements · Failure recovery design · Agentic UX patterns	LangChain agents or custom orchestration · Tool definitions and schemas · Reasoning loop patterns · LangSmith or similar tracing · Error recovery logic	Anthropic or OpenAI API · LangChain · Python or Node.js · LangSmith	Agent architecture diagram · Tool inventory · Observability spec · Failure recovery spec · PRD
	■ Closes Phase VII. Agentic AI is the current frontier. Placed here deliberately — after deep foundations.				

Graduate from product builder to platform thinker. GraphQL API design, cross-platform React Native mobile development, Docker containerization, full CI/CD pipeline automation, multi-tenant SaaS architecture, micro-frontend decomposition, production observability with SLA/SLO documentation, an executive reporting dashboard, an enterprise pricing migration as a GTM capstone, and a full SaaS product launch simulation as the flagship portfolio project. Portfolio milestone projects are assessed and designated during this phase — not before. PM skills introduced: platform and API strategy, mobile product management, infrastructure cost modeling, enterprise documentation, DevOps collaboration, incident response, DORA metrics, and end-to-end launch narrative construction.

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
	GraphQL API with Apollo Server				
P57	Replace or augment REST endpoints with GraphQL. Schema-first design, resolvers, mutations, real-time subscriptions, N+1 optimization via DataLoader, and Apollo Client integration on the frontend.	GraphQL vs REST decision documentation · Schema design as product contract · API versioning strategy · Query efficiency tradeoffs · Client-driven data fetching design	GraphQL · Apollo Server · Resolvers and mutations · DataLoader · Schema-first design · Apollo Client · Subscription handling	GraphQL · Apollo Server · Apollo Client · Node.js · TypeScript	API schema doc · REST vs GraphQL decision memo · Schema governance policy · PRD
	React Native Mobile App — iOS & Android				
P58	Port core product features to iOS and Android via React Native. Navigation, device APIs (camera, location), push notifications via APNs/FCM, offline mode with local storage, and app store submission prep.	Mobile-specific UX requirements · Platform parity decision matrix · App store submission spec · Offline-first product design · Mobile feature gap analysis vs web	React Native · Expo · React Navigation · Device APIs · APNs/FCM push notifications · TypeScript · Offline storage with MMKV	React Native · Expo · TypeScript · React Navigation · Firebase Cloud Messaging	Mobile PRD · Platform parity matrix · App store submission checklist · Push notification spec · PRD
	Docker Containerization & Docker Compose				
P59	Containerize the full-stack application. Multi-container Docker Compose setup: app server, PostgreSQL, Redis, and worker. Production-ready Dockerfile with multi-stage builds and environment variable management.	Infrastructure requirement writing · Environment parity documentation · Deployment risk log · Onboarding simplification via containerization	Docker · Docker Compose · Multi-stage builds · Health checks · Volume management · Environment variable injection	Docker · Docker Compose · Node.js · PostgreSQL · Redis	Deployment architecture diagram · Environment spec · Docker runbook · PRD
	CI/CD Pipeline with GitHub Actions				
P60	Full CI/CD pipeline: automated tests on every PR, build validation, staging deploy, manual production approval gate, rollback procedure documentation, and zero-downtime deploy confirmed.	Release process documentation · Deploy gate criteria · Rollback decision tree · Change management spec · DORA metrics awareness (deployment frequency, MTTR)	GitHub Actions workflow YAML · Test automation in CI · Environment secrets management · Blue-green deployment concept · Branch protection rules	GitHub Actions · Docker · Render or Railway or AWS	CI/CD pipeline diagram · Deploy runbook · Rollback procedure · Gate criteria doc · DORA baseline · PRD
	Multi-Tenant SaaS Architecture				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P61	Extend the product to support multiple organizations as tenants. Row-level security for complete data isolation, tenant-scoped API, org-level billing via Stripe, tenant onboarding flow, and tenant admin portal.	Multi-tenancy product scoping · Data isolation requirement writing · Tenant onboarding flow · Per-tenant pricing model · Enterprise sales readiness	PostgreSQL row-level security · Tenant context middleware · Shared schema with tenant_id · Stripe per-tenant billing · Tenant provisioning API	PostgreSQL · Node.js · Prisma · Stripe · React	Tenant architecture diagram · Data isolation spec · Onboarding flow · Tenant pricing model · PRD
	■ <i>Multi-tenancy is the architectural foundation of every B2B SaaS product.</i>				
	Micro-Frontend Architecture				
P62	Decompose the frontend into independently deployable micro-frontends via Webpack Module Federation. Shared design system, independent deployment pipelines per team, and cross-team ownership simulation.	Platform architecture documentation · Team topology design · API boundary definition · Cross-team dependency charting · Feature ownership models	Webpack Module Federation · Shared component library across MFEs · Shell application pattern · Independent CI/CD pipelines per module	Webpack Module Federation · React · TypeScript · GitHub Actions	Platform architecture diagram · Team boundary spec · Feature ownership matrix · PRD
	Observability — Logging, Metrics, Alerts & Incident Response				
P63	Production observability stack: structured logging, Sentry error tracking, performance metrics, uptime monitoring, on-call alerting, and a formal incident response runbook with post-mortem template.	SLA and SLO definition · Incident response process design · On-call runbook writing · DORA metrics (MTTR, deployment frequency) · Post-mortem culture	Structured logging (Winston or Pino) · Sentry error tracking · Uptime monitoring (Better Uptime or Pingdom) · Alert threshold design · PagerDuty or similar	Sentry · Datadog or Grafana · PagerDuty or similar · Node.js · GitHub Actions	SLA/SLO doc · Alert threshold spec · Incident runbook · Post-mortem template · PRD
	Executive Reporting & Business Intelligence Dashboard				
	★ PORTFOLIO CANDIDATE — Analytics PM: Executive Dashboard				
P64	Executive-level reporting: activation, retention, churn, MRR, NPS simulation, feature adoption, A/B experiment results summary, and custom SQL-driven report builder. CSV and PDF export.	Executive reporting design · Business intelligence fundamentals · Cohort analysis · Metrics hierarchy (leading vs lagging indicators) · KPI storytelling	Advanced SQL · Metabase or Redash · Custom charting · Cohort analysis queries · Report builder UI · CSV/PDF export	PostgreSQL · Metabase or Redash · Recharts · Node.js · React · SQL	Metrics hierarchy doc · Dashboard spec · SQL query library · Cohort analysis report · PRD
	Enterprise Usage Tracking & Consumption Pricing Migration				
	★ PORTFOLIO CANDIDATE — Cross-Functional GTM: Pricing Migration				

Proj	Scope	PM Growth	Tech Growth	New Tools	Artifacts
P65	Migrate from flat subscription to consumption-based pricing. Usage metering, real-time consumption telemetry, usage alerts, overage billing via Stripe, enterprise pricing negotiation tooling, and full GTM launch kit.	GTM launch kit construction · Internal alignment documentation (Engineering, Sales, Legal, Support) · Pricing migration risk log · Backward compatibility strategy · Post-launch feedback loop	Stripe usage-based billing · Real-time consumption telemetry logging · Usage metering middleware · Backward compatibility API strategy	Stripe Usage-Based Billing · PostgreSQL · Node.js · React · Segment.io	GTM launch kit · Pricing migration spec · Internal FAQ · Sales enablement update · Risk log · PRD
	Full SaaS Product Launch Simulation — Flagship Capstone				
	★ FLAGSHIP PORTFOLIO PROJECT — Full SaaS Capstone				
P66	Synthesize the strongest features into a cohesive production-grade SaaS: landing page, auth, onboarding, AI assistant, roadmap builder, feedback portal, subscription billing, analytics, admin, CI/CD, and complete GTM materials.	End-to-end product lifecycle ownership · Full case study suite construction · Launch narrative writing · Cross-functional artifact production · Business model documentation	Full-stack synthesis · All prior capabilities integrated · Production deployment · Live monitoring · Performance budget met · WCAG 2.2 AA confirmed	Next.js · TypeScript · PostgreSQL · Prisma · Stripe · OpenAI/Anthropic · Mixpanel · Segment · Resend · Vercel · GitHub Actions · Docker · Figma	Complete PMD artifact suite · Case study · Portfolio showcase entry · Launch retrospective
	■ <i>Threshold is earned, not scheduled. Coach assesses readiness from P55-P65 retrospective quality.</i>				

RECOMMENDED READING & OFFLINE ASSIGNMENTS — BY PHASE

Phase I–II Foundation & React	• Don't Make Me Think — Steve Krug · UX fundamentals every PM must internalize before building anything • The Pragmatic Programmer — Hunt & Thomas · Developer mindset and craft habits, essential for non-developers • HTML & CSS: Design and Build Websites — Jon Duckett · Visual illustrated entry point to the web • Learning React — Alex Banks & Eve Porcello · React mental model built from first principles
Phase II–III Backend & Auth	• Clean Code — Robert C. Martin · Writing code your future self and teammates will not curse • Designing Data-Intensive Applications — Kleppmann · The PM's database and distributed systems bible • The Phoenix Project — Kim, Behr, Spafford · DevOps and engineering culture through a compelling novel
Phase IV–V Analytics & Growth	• Lean Analytics — Croll & Yoskovitz · AARRR, cohort thinking, and the one metric that matters • Hacking Growth — Ellis & Brown · Systematic growth experimentation from the people who invented it • Trustworthy Online Controlled Experiments — Kohavi, Tang, Xu · A/B testing done rigorously • The Mom Test — Rob Fitzpatrick · How to talk to users without getting lied to
Phase IV–V Product Strategy	• Inspired — Marty Cagan · How the best product teams in the world actually work • Continuous Discovery Habits — Teresa Torres · Opportunity trees and user interview discipline • Competing Against Luck — Christensen · Jobs-to-be-Done theory at full depth • Obviously Awesome — April Dunford · Positioning as a PM discipline — what category are you in?
Phase VI–VII AI, Platform & Scale	• Building Machine Learning Powered Applications — Emmanuel Ameisen · ML from prototype to production • Platform Revolution — Parker, Van Alstyne, Choudary · Network effects and ecosystem design • The Lean Startup — Eric Ries · Build-measure-learn at every stage and every scale
Phase VII–VIII Leadership & Business	• Measure What Matters — John Doerr · OKRs in practice at Google and beyond • Crossing the Chasm — Geoffrey Moore · Technology adoption and the B2B PM challenge • High Output Management — Andrew Grove · Management fundamentals for PMs working with engineers • Thinking in Bets — Annie Duke · Decision-making under uncertainty — the PM's core operating condition
Offline Practice Throughout the Journey	• PM Exercises (pmexercises.com) · Daily product design and metric practice problems — do one per day • Lenny's Newsletter (lennyrachitsky.com) · Best practitioner-written PM content available today • Stratechery (stratechery.com) · Ben Thompson's business and technology strategy — read every piece • Monthly product teardown habit: pick one app per month, write a 500-word friction audit • User interview practice: conduct 2 real user interviews per project using The Mom Test methodology • SQL practice: Mode Analytics SQL Tutorial · 30 minutes per day during Phases III through V

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